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REVIEW | *Integrative Cardiovascular Physiology and Pathophysiology*

Molecular effects of exercise training in patients with cardiovascular disease: focus on skeletal muscle, endothelium, and myocardium

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Adams V, Reich B, Uhlemann M, Niebauer J. Molecular effects of exercise training in patients with cardiovascular disease: focus on skeletal muscle, endothelium, and myocardium. *Am J Physiol Heart Circ Physiol* 313: H72–H88, 2017. First published May 5, 2017; doi:10.1152/ajpheart.00470.2016.—For decades, we have known that exercise training exerts beneficial effects on the human body, and clear evidence is available that a higher fitness level is associated with a lower incidence of suffering premature cardiovascular death. Despite this knowledge, it took some time to also incorporate physical exercise training into the treatment plan for patients with cardiovascular disease (CVD). In recent years, in addition to continuous exercise training, further training modalities such as high-intensity interval training and pyramid training have been introduced for coronary artery disease patients. The beneficial effect for patients with CVD is clearly documented, and during the last years, we have also started to understand the molecular mechanisms occurring in the skeletal muscle (limb muscle and diaphragm) and endothelium, two systems contributing to exercise intolerance in these patients. In the present review, we describe the effects of the different training modalities in CVD and summarize the molecular effects mainly in the skeletal muscle and cardiovascular system.

physical activity; shear stress; vessel; atherosclerosis; heart failure

NO OTHER DISEASE affects Western societies more in terms of morbidity, mortality, and disease-related costs than cardiovascular disease (CVD), including coronary artery disease (CAD) and chronic heart failure (CHF). Fatty and high-caloric nutrition, an unfavorable lipid profile, hypertension, smoking, and physical inactivity are associated with a higher prevalence of CVD (85, 98, 102). For decades, physicians followed the dogma of "rest and pain" and advocated prolonged rest for the majority of their patients (187). This dogma changed after the publication by Morris et al. investigating the incidence of CAD in London bus drivers and conductors (148). The result clearly documented that the more active conductors exhibited a far lower incidence of CAD compared with the age-matched physically less active bus drivers. This relation between fitness and the incidence of CVD was confirmed by several investigators studying >100,000 individuals [reviewed by Lee et al. (120)]. Despite this knowledge of the benefits of regular exercise, too many individuals are still exercising below the recommendations (109). On the other hand, many who are already physically active are on the search for exercise training programs of maximal gain with minimal pain. To attract sedentary subjects or to improve compliance with lifelong training prescription of those who have already embarked on this endeavor, an increasing number of novel and sometimes innovative training programs have emerged (105). Such a strategy is of utmost interest, as habitual physical activity is not only effective in the prevention and treatment of CVDs but also other leading noncommunicable diseases such as type 2 diabetes (107), osteoporosis (75), obesity (143), depression (209), cognition (142), and breast cancer (181).

Exercise training exerts its beneficial effects via different molecular mechanisms, which result in attenuated progression as well as regression of coronary atherosclerosis and induce further beneficial effects in different systems in the human body, e.g., the myocardial, cardiovascular, muscular, metabolic, neural, respiratory, and thermoregulatory systems (5, 31, 157, 158, 183, 224).

In this review, we will briefly describe the most common modes of exercise training [i.e., continuous endurance training (CET), high-intensity interval training (HIIT), pyramid training (PYR), and resistance training (RT)] and discuss how exercise training works at the molecular level in different organ systems (skeletal muscle, endothelium, and myocardium).

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Different Training Modalities

CET. CET is a type of physical activity that is performed over a longer period of time at a constant, mostly submaximal workload of sufficient intensity to increase heart rate (HR) and blood pressure (Fig. 1*A*). The efficacy of CET has been well documented (189). As a matter of fact, CET is the most common type of exercise during daily life since it includes, e.g., (Nordic) walking, hiking, running, biking, cross-country skiing, and swimming. Therefore, most retrospective as well as epidemiological studies that have assessed the effects of daily physical activity unavoidably study the effects of CET, which is why the bulk of scientific literature stems from there and why most recommendations favor this kind of exercise. CET reliably and reproducibly improves peak work capacity (PWC) and has been shown to be effective in reducing cardiac as well as all-cause mortality (38, 123, 200). In stable CAD patients, 3×1-h sessions of CET at intensities of around 75–80% of peak HR over 6 wk already improved peak work capacity by 21% (207).

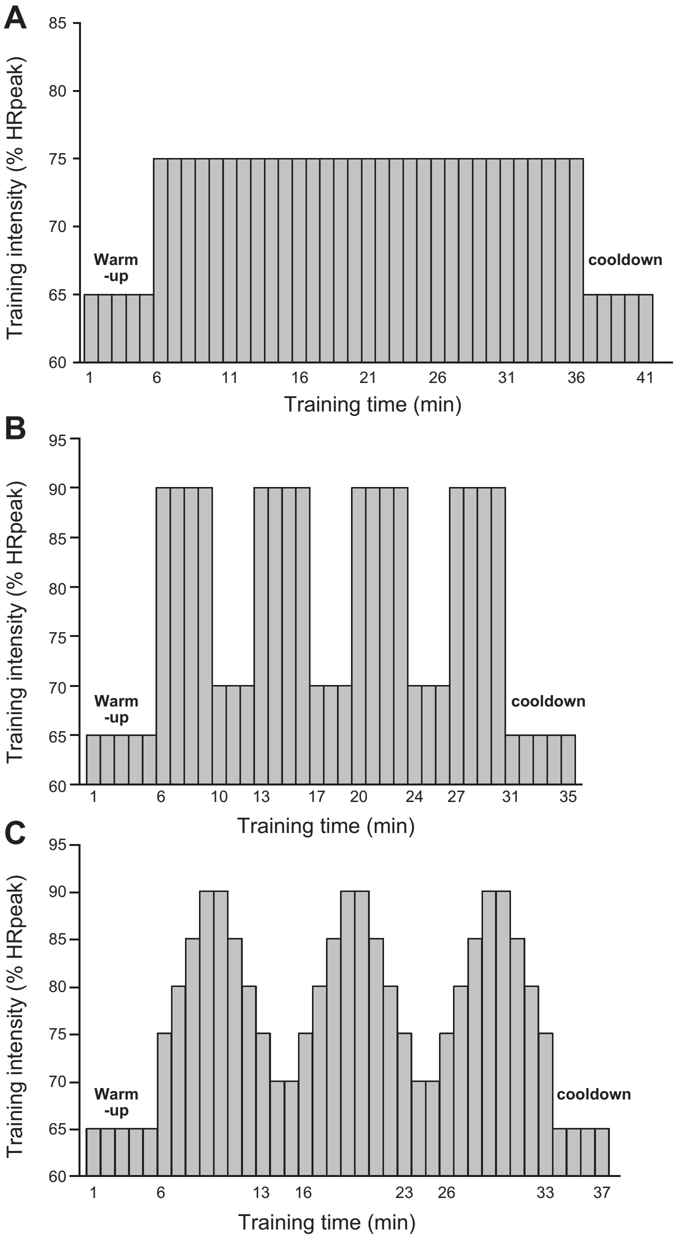

Fig. 1. Schematic drawing of different training protocols: moderate continuous exercise training (*A*), high-intensity interval training (*B*), and pyramid training (*C*) with respect to time and exercise intensity measured as percent of heart rate peak.

Already in the early 1980s, Ehsani and colleagues studied intense exercise training [intensity ranged from 60 to 70% of peak O_2 uptake ($\dot{V}O_{2peak}$) for the first 3 mo followed by 70−90% of $\dot{V}O_{2peak}$ for the remaining period] in patients with CAD (57, 58, 170). They reported a reduction in myocardial ischemia and that the beneficial effect could be maintained if the patients continued to exercise. Also, in CHF, >20 studies have shown that CET is effective in patients with impaired left ventricular function, where improvement of 18–25% $\dot{V}O_{2peak}$ or 18–34% in peak exercise duration could be attained (17). In one of the largest exercise heart failure (HF) trials, the HF-Action trial, an improvement in exercise capacity was detected, but in the protocol-specified primary analysis, exercise training resulted in nonsignificant reductions in the primary end point of all-cause mortality or hospitalization. However, after pre-specified adjustment for highly prognostic predictors of the primary end point was performed, exercise training was associated with modest significant reductions for both all-cause mortality or hospitalization and cardiovascular mortality or HF hospitalization (161). Also, when correcting for compliance, the mortality rate was lower in the exercise-trained group, and a relation between the volume of exercise and clinical outcome was observed (106). This result could be confirmed by Belardinelli et al. (24), who performed a CET program in HF patients for >10 yr with very high compliance. In that study, moderate supervised exercise training performed twice weekly for 10 yr maintained functional capacity and conferred a sustained improvement in quality of life. This was associated with reduction in major cardiovascular events, including hospitalizations for CHF and cardiac mortality. In recent years, the treatment of HF patients with still preserved ejection fraction (HFpEF) by exercise training attracts attention, because pharmaceutical treatment shows only limited benefits in this patient group (40, 55, 186, 192, 233). Preliminary results from small monocenter clinical trials have suggested that exercise training increases exercise capacity and improves quality of life [summarized by Palau et al. (162)]. Nevertheless, large multicenter trials, such as the OptimEx trial (195), which is ongoing, are warranted to clarify the role of exercise training as a therapy for HFpEF.

As there are not only physiological factors responsible for improved exercise capacity but also psychological attributes, the explanation for improvement in physical capacity is very complex. Besides improving PWC, CET can also help to significantly reduce hopelessness, depression, and suicide ideation as well as improve sense of belonging in high-risk suicide patients (160, 209). This is also important in patients with CAD, as they often have to cope with challenging psychological situations, which make them prone to an unhealthy lifestyle including a sedentary lifestyle.

Guidelines for the prevention and rehabilitation of CVDs from leading professional international societies encourage healthy subjects and patients alike to include aerobic CET into their daily lives. In their recommendations, the European Society of Cardiology favors moderate to vigorous intensities at a duration of 30 min/session at 3–5 days/wk, totaling at least 150 min/wk (167a), whereas the American College of Sports Medicine and the American Heart Association are more precise and recommend aerobic CET on at least 3 days/wk for 20–60 min/session plus a 10- to 15-min warmup and cooldown at intensities between 40 and 80% HR reserve (84). Despite minor differences, all recommendations have in common that CAD patients should train 3–5 times/wk for 30–60 min/day, with a minimum duration per week of 150 min at an intensity of 70% maximum HR (HR_{max}).

In summary, CET is the preferred type of endurance exercise in all guidelines of the leading professional societies, both for the bulk of evidence as well as its simplicity. With minimal instructions, CET can be performed by anybody who is physically able to exercise.

HIIT. HIIT is widely used in competitive sports to improve PWC and has recently been validated in patients with CHF (226). HIIT is characterized by brief intermittent bursts of exercise interspersed by active recovery periods (Fig. 1*B*). For example, Helgerud et al. (88) introduced 4 × 4-min intervals at 90–95% HR_{max} with 3-min active recovery periods at 70% HR_{max} between each interval. The HIIT protocols used in several studies in CAD or CHF differ with respect to the definition of "high-intensity interval." Definitions such as 90–95% peak HR (42, 146, 171), 80–90% peak HR (105), 90% $\dot{V}O_2$ reserve (218), or 80–104% peak power output (PPO) (46) have been reported. Also, the duration and number of repetitions of the maximal load vary between studies. Currie et al. (46) used a protocol of 10 × 1-min intervals at intensities of 89% of PPO with 1 min at 10% PPO during the resting phase, whereas in the studies from the Trondheim group (146, 171), four intervals for 4 min at 90–95% peak HR were performed.

Since the introduction of HIIT by Wisløff et al. in 2007 (226), several studies have been performed using HIIT in various patient populations (42, 59, 145, 164). For example, the recently published EXCITE trial (145) evaluated the impact of HIIT on coronary collateral blood flow (measurement for collateral recruitment/formation). The authors reported a significant increase in coronary collateral blood flow in response to both moderate- and high-intensity exercise training with no difference between training intensities. In recent years, several clinical studies performed in either CAD or CHF patients also confirmed an improvement of $\dot{V}O_{2peak}$ by HIIT [for summaries, see Haykowsky et al. (87) and Liou et al. (127)].

PYR. PYR, characterized by a stepwise increase and decrease of work load (Fig. 1*C*), is another promising training protocol widely used in competitive sports (92). Just recently, our group examined the effects of 3 × 8 min of stepwise load increase and subsequent decrease from 65% to 95% to 65% peak HR in patients with stable CAD (207). We found that PYR was as effective as HIIT and CET with regard to improvement in PWC and that it can be performed with very well comparable mean HRs and energy expenditure. PYR might be an attractive alternative for those looking for short, time-saving training programs during cardiac rehabilitation or home-based exercise training, because the results are comparable to CET. Therefore, we suggested that a combination of the three training programs can be offered on individual preference, helping to improve adherence to lifelong behavioral changes.

CET Versus HIIT: which should be preferred? As mentioned above, HIIT and CET improve $\dot{V}O_{2peak}$, but which method is more effective and should be preferred for CAD and CHF? For CAD, this question was recently addressed by two meta-analyses (60, 127). In both studies, HIIT seemed to be superior compared with CET with respect to the improvement in exercise capacity ($\dot{V}O_{2peak}$). Overall, HIIT was associated with a rise in $\dot{V}O_{2peak}$ of 1.78 ml·kg^{-1}·min^{-1} (127) or 1.53 ml·kg^{-1}·min^{-1} (60). Nevertheless, the authors still concluded that long-term studies are necessary to investigate morbidity and mortality after HIIT. This conclusion is also supported by a recent review by Hussain et al. (95).

With respect to CHF, the evidence of superiority of CET or HIIT is less clear. The first study applying a HIIT program to patients with CHF was performed by Wisløff et al. (226). In their study, a superior cardiovascular effect of HIIT was seen compared with CET. Also, a recently published meta-analysis including 7 unique trails with 168 patients (80 HIIT and 88 CET) concluded that HIIT is more effective than CET for improving $\dot{V}O_{2peak}$ but not left ventricular ejection fraction (LVEF) (87). However, it must be considered that four of the seven studies reported no difference between HIIT and CET (54, 96, 152, 190). This notion is an agreement with the result of the recently published SmartEx study, a multicenter study with 261 CHF patients randomized into either a control, a CET, or a HIIT group (59). With respect to changes in LVEF, $\dot{V}O_{2peak}$, and left ventricular end-diastolic diameter, no difference was found between CET and HIIT. Therefore, the superiority of HIIT in CHF is still questionable, and its feasibility in CHF remains unresolved. Nevertheless, we should recognize that according to a recent meta-analysis performed in CHF patients, training frequency and session duration yielded the largest improvement in exercise capacity (216).

RT. In recent years, RT has been included into the training therapy of patients with CAD and CHF [reviewed by Cornish et al. (43) and Giuliano et al. (73)]. This is based on the fact that RT improves muscle strength in middle-aged (64, 103) and elderly patients (32) and that insufficient isokinetic strength of extensor and flexor muscles had a strong negative association between peak torque and the severity of disease in HF patients (94). As a consequence, professional medical societies recommend hypertrophy resistance training in their current guidelines (12, 70, 137). Improvement in muscle mass and strength between 2% and 15% in women and men at all ages can be achieved [summarized by Egger et al. (56), Stewart et al. (194), and Van Der Heijden et al. (211)] and is associated with improvement in bone density, body fat percentage, fat-free mass, insulin response to glucose stimulation, basal insulin levels, insulin sensitivity, and blood pressure.

Analyzing the effects of RT on muscle strength, exercise capacity, and mobility in CAD patients, a recently published meta-analysis (234) concluded that RT in combination with aerobic training increased exercise capacity and muscle strength in middle-aged and elderly patients and mobility in the elderly cohort. In addition, combined strength-endurance training seems to induce more pronounced increases in muscle strength and muscle mass compared with steady-state endurance training (37, 133, 139). With respect to CHF, a recently published meta-analysis, including 10 studies and 240 participants, concluded that RT is able to increase muscle strength, aerobic capacity, and quality of life, thereby offering an alternative approach, particularly for those patients unable to participate in aerobic training (73).

Molecular Changes in Different Organs Induced by Exercise Training

Skeletal muscle. Skeletal muscle represents the largest pool of proteins in the organism, and its proper function is essential for human functioning with respect to locomotion and breathing. Therefore, a tight regulation between factors influencing protein synthesis and degradation is important to control muscle mass. Loss of skeletal muscle mass directly contributes to exercise intolerance and impaired daily activities, which is a strong predictor of quality of life and mortality (13, 236). Besides the alterations in skeletal muscle mass, secondary modification of contractile proteins (ubiquitinylation and carbonylation) also leads to impairment of contractile force generation (41). To circumvent muscle wasting and reduced force generation, numerous interventions such as pharmacological and nutritional aids have been used, but with limited success (63). An alternative clinical proven intervention is exercise training, providing the most benefits at the molecular and physiological level. In the following paragraphs, molecular changes in the skeletal muscle occurring in CVD and the impact of exercise training will be discussed.

LIMB MUSCLE. *Muscle growth and atrophy*. As mentioned above, a balance between anabolic and catabolic factors is crucial for maintaining muscle mass. Muscle loss in skeletal muscle is mainly driven by the ubiquitin-proteasome system (UPS) (16, 31, 149), autophagy (34, 175), and myostatin-mediated signaling (169, 180).

In the UPS, three enzyme systems trigger protein degradation: ubiquitin-activating enzymes (E1), ubiquitin-conjugating enzymes (E2), and ubiquitin ligases (E3) (199a). Although many E3 ligases are known, muscle ring finger 1 (MuRF1) and muscle atrophy F-box (MAFbx/atrogin) are ubiquitin E3 ligases specifically expressed in skeletal muscle (27). They exhibit increased expression in models of muscle atrophy (27, 39, 72, 74, 115) and are considered important markers for muscle atrophy. This notion is further supported by the observation that mice deficient in these E3 ligases are resistant to muscle atrophy (27) and that MuRF1 seems to be essential for TNF-α-induced loss of muscle function (4). What are targets for MuRF1- and atrogin-mediated proteolysis? With the use of yeast two-hybrid screens of skeletal muscle cDNA libraries with MuRF1 baits, eight potential myocellular targets of MuRF1-dependent ubiquitination were identified: titin, nebulin, the nebulin-related protein (NRAP), troponin I, troponin T, myosin light chain 2, myotilin, and titin-cap (T-cap) (227). With respect to targets of MAFbx, myoblast determination protein (MyoD), an important factor regulating myoblast identity and differentiation, was identified by yeast two-hybrid screens (204). In addition, overexpression of a mutant MyoD-Lys133Arg lacking MAFbx-mediated ubiquitination prevented atrophy of mouse primary myotubes and skeletal muscle fibers (110). Depending on the site of ubiquitination (Lys^{48} or Lys^{63}) and the length of the attached ubiquitin moieties, different pathways are activated (Fig. 2). Taken together, the activation of these two ubiquitin E3 ligases modulates the concentration of contractile proteins and muscle differentiation. Analyzing skeletal muscle specimens obtained from either patients or animal models with CHF or CAD revealed an increased expression of the atrogenes *MuRF1* and *MAFbx* (20, 72, 135, 163).

Myostatin, a member of the transforming growth factor-β family, is a negative regulator of muscle mass. It is produced and released by myocytes, and by autocrine mechanisms it modulates muscle growth and differentiation (116). Animals

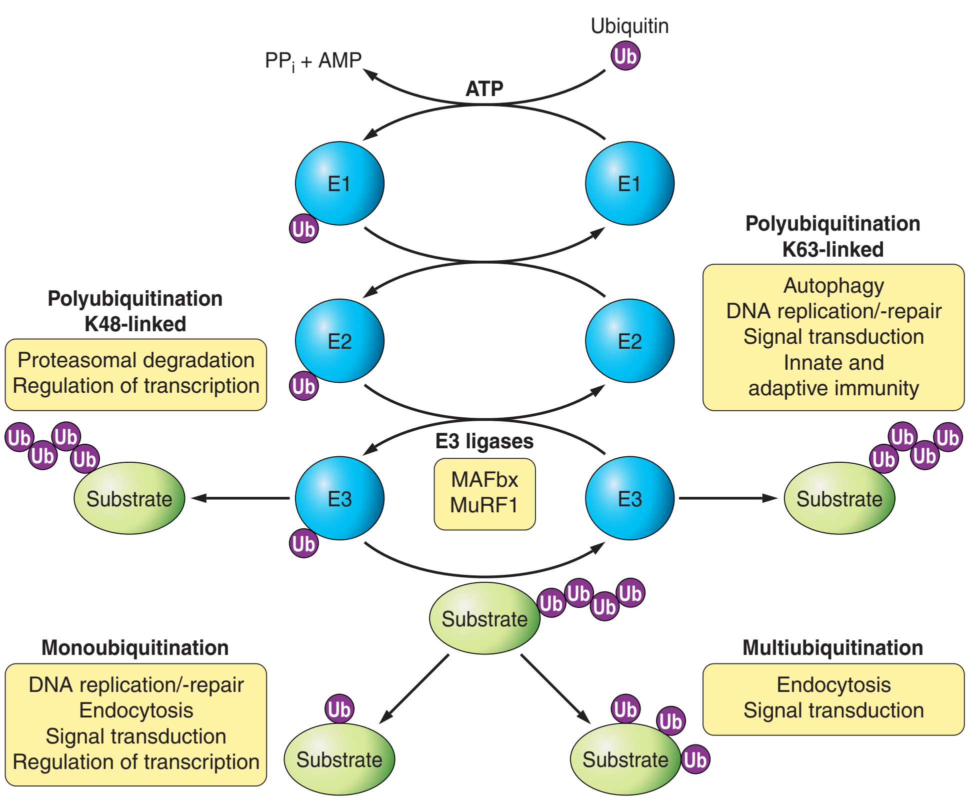

Fig. 2. Schematic drawing of the protein ubiquitination cascade. Three enzymatic systems are involved in attaching ubiquitin moieties to proteins: ubiquitin-activating enzymes (E1), ubiquitin-conjugating enzymes (E2), and ubiquitin ligases (E3) such as muscle ring finger 1 (MuRF1) and muscle atrophy F-box (MAFbx). Depending on the site of the ubiquitin attachment (Lys^{48} or Lys^{63}) and the length of the attached ubiquitin chain, different pathways are activated. PP_i, inorganic pyrophosphate.

lacking the myostatin gene (44, 141) or animals treated with substances inhibiting myostatin (151) exhibit greater muscle mass. Even individuals who have mutations in both copies of the myostatin gene have significantly more muscle mass and are stronger than normal (182). Mature myostatin binds to the activin receptor in skeletal muscle and activates Smad2 and Smad3 (177) via phosphorylation, thereby inhibiting Akt/target of rapamycin complex 1 (TORC1)/p70S6K signaling and muscle growth (206). When skeletal muscle samples from either heart failure patients (121) or animal models with CHF were analyzed (122), the expression of myostatin was significantly increased.

Autophagy, a process where targeted protein or even organelles are trapped in autophagosome before the content is degraded in lysosomes, is elevated in muscle-wasting conditions such as denervation in aging (161a), fasting (131), and CHF (99).

Does exercise training influence the UPS, autophagy, and myostatin system in CVD? According to current literature, there is clear evidence that exercise training modulates the UPS and myostatin system. In a mouse model of sympathetic hyperactivity-induced HF, CET (4 wk, 5 days/wk at 60% of maximal work load) reduced MuRF1 and MAFbx mRNA expression and normalized the concentration of ubiquitinylated proteins and proteasome activity (45). In a further animal study, Souza et al. (192a) investigated the impact of CET (10 wk, 5 days/week for up to 22 min at the lactate threshold) performed during the transition from cardiac dysfunction to HF on the expression of anabolic and catabolic factors, hypothesizing that skeletal muscle wasting could be prevented. They reported that the expression of MuRF1 and MAFbx was reduced compared with the nonexercising control group. In addition, muscle wasting was also reduced. This impact of exercise training on the UPS has also been observed in humans (72). In the Leipzig Exercise Intervention in Chronic Heart Failure and Aging catabolism study, Gielen and colleagues (72) reported, after 4 wk of CET (training at 70% $\dot{V}O_{2peak}$), a reduction of MuRF-1 mRNA expression by -32.8% in patients aged $\leq$55 yr and by -37.0% in patients aged $\geq$65 yr. This impact of exercise training on the UPS could also be confirmed in HF patients in New York Heart Association (NYHA) class IIIb (91).

With respect to myostatin, only one study in CHF patients examining the impact of CET on myostatin expression in the skeletal muscle is available. Lenk et al. (121) investigated 24 CHF patients (stage NYHA IIIb) randomized to either a sedentary control group or a CET group (12 wk, daily at 60% peak HR). Exercise training led to a 36% reduction of the mRNA and a 23% decrease of the myostatin protein compared with baseline.

Besides increased protein degradation, reduced protein synthesis induced by anabolic factors may also impact on muscle mass. One of the most potent anabolic factors is insulin-like growth factor I (IGF-I), which has been shown to predict altered body composition, anabolic deficiency, as well as cytokine and neurohumoral activation in CHF patients (155). The potency of IGF-I to modulate muscle mass is evident most impressively in IGF-I transgenic animals, where localized IGF-I transgene expression in the skeletal muscle resulted in increased muscle mass mainly in muscles rich in fast fibers, which was associated with increased force generation (150). Furthermore, it even prevented skeletal muscle wasting normally observed in CHF (184). Only a few studies have investigated the impact of CET on the expression of IGF-I in the skeletal muscle. Muscle biopsies from stable CAD patients were obtained before and after 8 wk of low-intensity concentric and eccentric endurance-type training, and analysis of several factors including IGF-I showed that IGF-I mRNA levels were significantly increased (237). Also, in CHF patients, an exercise-training program over a period of 6 mo resulted in an 81% increase in IGF-I expression in skeletal muscle.

Modulation of contractile proteins by ROS. Myofibrillar proteins are highly susceptible to oxidative damage both in vitro and in vivo, and recent data have provided evidence that oxidation of sarcomeric proteins contributes to reduced force production (41). A variety of oxidative modifications of sarcomeric proteins can occur including carbonylation of lysine, arginine, and proline residues, nitration of tyrosine residues, thiol oxidation, and disulfide bond formation (191). Potential sources for an increased load of ROS are NAD(P)H oxidase, xanthine oxidase (XO), and the mitochondria through uncoupling of the electron transport chain and the leak of electrons from complex I and complex II (165; see below). In skeletal muscle samples obtained from either experimental models of CHF (21, 30) or patients with CHF (125), increased protein expression/enzymatic activity of NAD(P)H oxidase and XO was evident. Does exercise training have an impact on ROS-mediated protein modification? When human muscle biopsies obtained from CHF patients randomized either to a sedentary or to a CET group (6 mo, 20 min/day at a HR that reached at 70% of maximum O_2 uptake) were analyzed, nitrotyrosine modulation of proteins was reduced by 35% (125). Also, in an experimental setting of CHF (left anterior descending coronary artery ligation in mice), CET reduced the amount of carbonylated proteins (45).

Modulation of mitochondrial energy supply. The regeneration of phosphocreatine by mitochondrial ATP is essential for long-lasting muscle contraction (26). Therefore, tight control of mitochondrial function in skeletal muscle cells is crucial. Several lines of evidence exist showing that in CHF and CAD, mitochondrial biogenesis as well as mitochondrial ATP production is impaired [reviewed by Rosca and Hoppel (173) and Ventura-Clapier et al. (214)].

With respect to exercise training and its impact on mitochondrial function, several experimental (30, 33) and human studies (23, 78, 176, 205, 223) are available proving beneficial effects. In general, CET resulted in an augmented ATP production with a direct correlation to increases in $\dot{V}O_{2peak}$ (206), volume (23), and surface density of mitochondria (78) as well as improved mitochondrial respiration and a better respiratory control ratio (30).

Besides the direct effects on mitochondria, another intriguing fact is that exercise training increases nitric oxide (NO) production in skeletal muscle by stimulating the neuronal isoform of NO synthase, which stimulates mitochondrial biogenesis through AMP-activated protein kinase and an upregulation of peroxisome proliferator-activated receptor-γ coactivator-1α (PGC-1α) (178).

Modulation of microRNAs. As documented above, the skeletal muscle is an organ with great plasticity and is influenced by exercise training in healthy and diseased individuals. MicroRNAs (miRNAs) specifically expressed in the skeletal muscle are called "myomirs," and miR-1, miR-133a, miR-133b, and miR-206 represent ~25% of all miRNAs expressed in the skeletal muscle (185). Nielsen et al. (159) provided data in healthy young men that myomir expression transiently increased 1 h after a single bout of exercise, whereas myomir expression at rest decreased to a new steady-state level after 12 wk of aerobic training, returning to preexercise levels after a period of inactivity. So far, no studies on the influence of exercise training on myomir expression in CAD or CHF are available. Besides these classical myomirs, other miRNAs are also influenced by physical activity, such as miR-696 (15). Microarray analysis of mouse skeletal muscle subjected to either exercise training or inactivity (hind limb suspension) identified miR-696 as a physical activity-dependent miRNA (downregulated by exercise and upregulated by inactivity) regulating the expression of peroxisome PGC-1α (15), thereby promoting mitochondrial biogenesis and activation of respiratory enzymes (229). With respect to RT, decreased expression of miR-378 was noted in the vastus lateralis after 12 wk (47). In addition, a change in miR-378 level with RT is correlated with gain in lean body mass. In that respective study, no change was detected for the classical myomirs.

In summary, some data are available documenting the influence of exercise training on the expression of miRNAs in the skeletal muscle, but data for specific patient groups such as CAD or CHF are still missing.

DIAPHRAGM. Dyspnea frequently limits daily activities in patients with CHF, and one potential cause is inspiratory muscle dysfunction mainly of the diaphragm. Respiratory muscle dysfunction is now recognized as a silent feature of patients (51, 81, 82, 132, 140) and animals (30, 41, 93, 119, 197) exhibiting systolic or diastolic heart failure. At the molecular level, several mechanisms leading to diaphragm dysfunction are currently being investigated. A shift in fiber type composition, as already documented for the limb muscles, could play a role. Indeed, a shift from type II (very high force production) to type I (low force production) muscle fibers was observed in animals with diastolic HF (30) and biopsies obtained from CHF patients (202, 203), which is in contrast to the peripheral limb muscle, where a shift from type I to type II was reported (52). Besides alterations in fiber type composition, fiber atrophy (30, 193), with an upregulation of atrophy markers such as MuRF1, MAFbx, and caspase (189, 191), a decreased Ca^{2+} sensitivity of force generation (198), and a reduced myosin heavy chain content per half sarcomer (212), could also be noted. Another potential mechanism responsible for reduced force generation of the diaphragm is the carbonylation of contractile proteins (mainly actin and myosin heavy chain), thereby impairing proper function (29, 41). The increased carbonylation and the subsequent loss of function are due to increased load of ROS. In an elegant study using p47*phox* knockout animals, Ahn et al. (6) convincingly documented the importance of NAD(P)H oxidase for modulating diaphragm contractile force. Another source of ROS leading to contractile dysfunction is the mitochondria, because CHF animals treated with a mitochondria-targeted antioxidant showed normal diaphragm function (111). Taken together, several pathways activated by myocardial damage lead to diaphragm muscle weakness (Fig. 3).

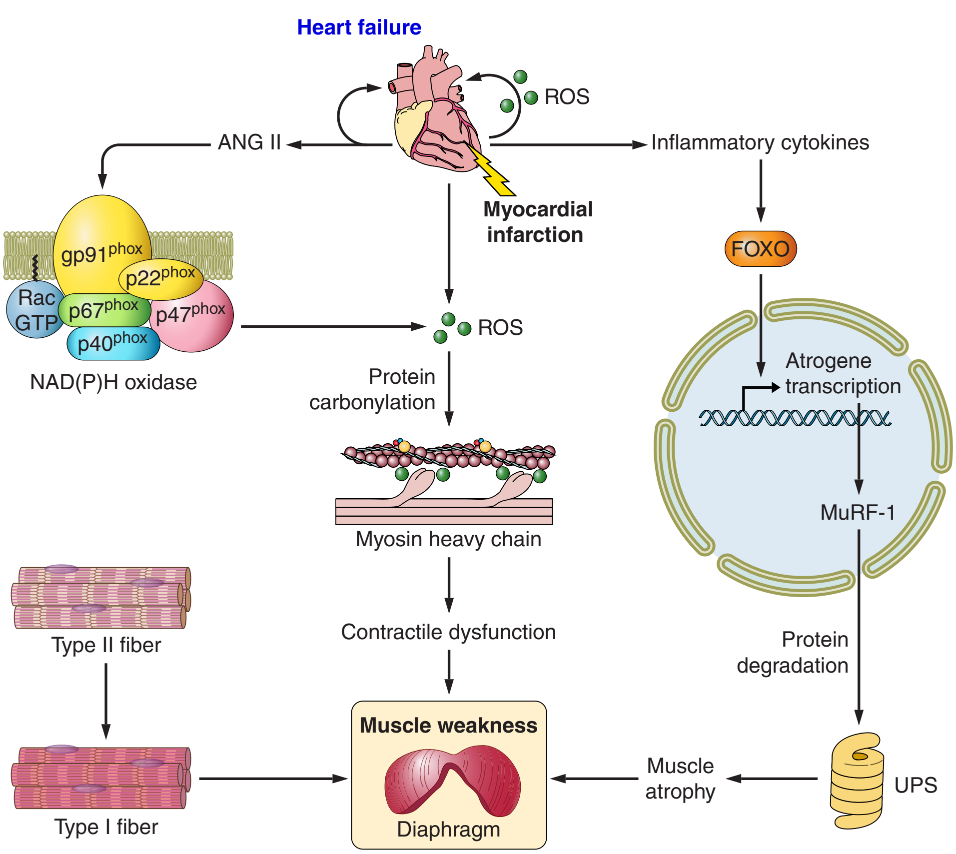

Fig. 3. Summary of potential mechanisms involved in respiratory (diaphragm) muscle weakness in heart failure patients, which contributes to more severe symptoms (i.e., breathlessness and fatigue), a reduced quality of life, and an increased mortality. Cardiac dysfunction is known to mediate a cascade that leads to diaphragm dysfunction via paracrine and autocrine mechanisms that result in contractile dysfunction and muscle atrophy. This may be underpinned by an increase in ROS from NADPH oxidase and xanthine oxidase, an increase in protein degradation by MuRF1, and a fiber type shift.

With respect to the impact of aerobic exercise training on diaphragmatic alterations, only a few studies are available (30, 50, 134). In an experimental setting (134), CET of mice on a treadmill prevented the TNF-α-induced force reduction and led to an enhanced mRNA expression and activity of glutathione peroxidase. Carbonylation of proteins, in particular of α-actin and creatine kinase, was diminished by exercise training. Also, in an animal model of CHF with preserved ejection fraction, HIIT over a period of 28 wk prevented mitochondrial and functional impairments of the diaphragm (30). In addition, exercise training also impacts on the integrity of neuromuscular junctions and the expression of nicotinic ACh receptors (50).

Endothelium. It has been said that "one is as old as one's arteries" or, a little bit modified, that "one is as old as one's endothelium" (11). Several reasons can be given as to why endothelial research has become relevant for today's mechanistic concepts in understanding the beneficial effect of exercise training. First, endothelial dysfunction has been recognized as a hallmark for the development of atherosclerosis (174), and therefore the prevention of endothelial dysfunction by exercise training will prevent the development of atherosclerosis (154). Second, the occurrence of endothelial dysfunction is a strong predictor for future cardiovascular events (167, 179), and therefore it is often used in clinical studies as a surrogate end point (113). Finally, a large variety of methods are available measuring endothelial dysfunction in humans [reviewed by Higashi (90) and Vizzardi et al. (215)] as well as in the experimental setting (3, 19).

The clinical effects of CET on coronary endothelial function were first studied in humans by Hambrecht et al. (80). They reported that a 4-wk CET program was effective in attenuating the paradoxical arterial vasoconstriction in epicardial conduit vessels by −54% and increased average peak flow velocity by +78%. Subsequently, the improvement in endothelial function by endurance exercise training was confirmed by several studies in patients with CAD (10, 22, 76, 130) or diabetes (53, 188). More recently, a substudy of the Study on Aerobic Interval Exercise Training in CAD (SAINTEX-CAD) compared the impact of CET training with a HIIT training protocol on endothelial function (210). In this large multicenter study, both training regimens improved endothelial function without a difference between both training formats. This is in contrast to the earlier small monocenter study by Wisløff et al. (226), where a superior effect of HIIT on endothelial function was reported. With respect to the molecular mechanisms responsible for the beneficial effect of exercise training on endothelial function, several mechanisms besides the modulation of endothelial damage (8) have to be discussed.

EXERCISE AND ACTIVATION OF ENDOTHELIAL NO SYNTHASE. The production of NO, the main factor for regulating vasodilation, is mediated by endothelial NO synthase (eNOS). The activation of eNOS relies on a multistep activation. The first step in this activation cascade is Ca^{2+} dependent. The entry of Ca^{2+} into the cell and its interaction with calmodulin trigger the release of eNOS from plasma membranes, where it is attached and inactive to caveolin-1. As a second step, eNOS associates with heat shock protein (Hsp)90 to be protected from degradation and leading to the dimerization of the enzyme, which is the activated form of the enzyme producing NO (18, 35, 66). Hsp90 also helps to recruit kinases and phosphatases, further activating the enzyme (67). Upon increasing laminar shear stress by exercise training, the signaling cascade is initiated via activation of mechanical receptors, such as VEGF receptor 2 (100) or platelet endothelial cell adhesion molecule 1 (PECAM1) (68) followed by phosphatidylinositol 3-kinase (PI3K)/Akt-dependent phosphorylation of eNOS at Ser^{1177} (Fig. 4). Another pathway of shear stress-mediated eNOS activation is via integrins, linking the extracellular matrix with the actin network. Integrins interact with different molecules

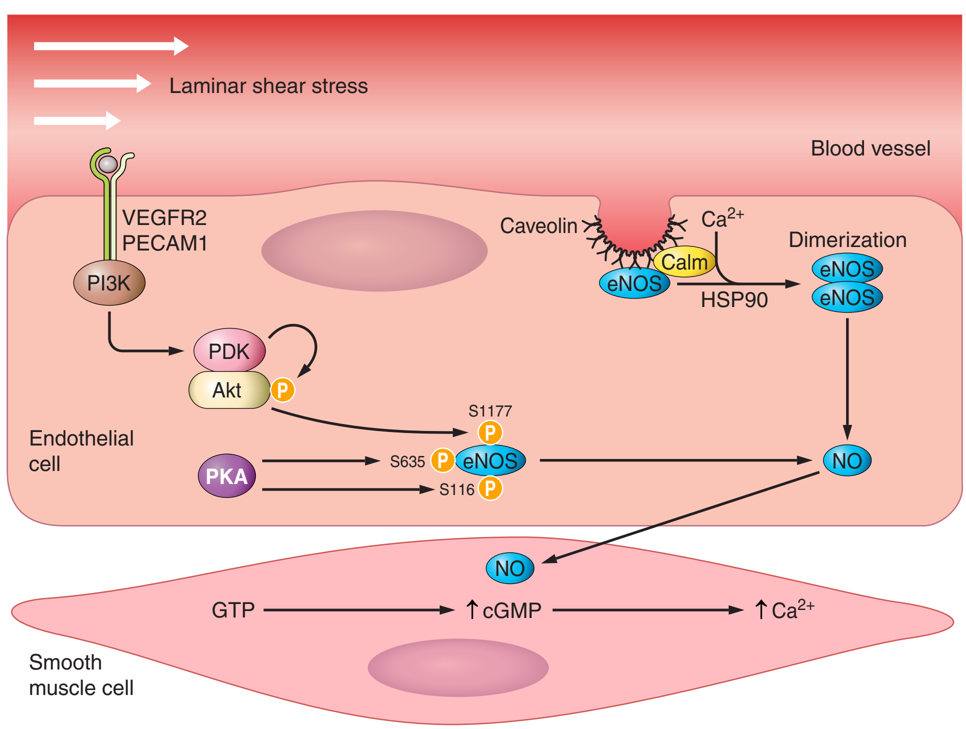

Fig. 4. The majority of exercise effects on the vascular endothelium are mediated by intermittent increases of laminar shear stress. On the luminal side of the endothelial cells, direct signaling can occur through deformation of VEGF receptor 2 (VEGFR2) or platelet endothelial cell adhesion molecule 1 (PECAM1) activating phosphatidylinositol 3-kinase (PI3K) to phosphorylate Akt (PKB) and induce Akt-mediated endothelial nitric oxide (NO) synthase (eNOS) phosphorylation, leading to higher NO production. PDK, phosphoinositide-dependent kinase.

involved in signaling, such as focal adhesion kinase (FAK) Src kinase, Fyn kinase, or p130 (124). For example, the pivotal role of FAK activation for NO-dependent vasodilation was documented using specific phospho-FAK antibodies, thereby preventing activation of Akt and eNOS phosphorylation (108). Besides these experimental data, the exercise training-dependent modulation of Akt and eNOS expression was also investigated in patients with CAD (76) and CHF (62) with diverging results. Ennezat et al. (62) reported no increased steady-state transcript levels for eNOS by exercise training, whereas in the study of Hambrecht et al. (76), patients in the training group had twofold higher eNOS protein expression and fourfold higher eNOS Ser^{1177} phosphorylation levels in the endothelium. Furthermore, a linear correlation between Akt phosphorylation and phospho-eNOS levels and between phospho-eNOS and the change in endothelial function was documented.

EXERCISE AND OXIDATIVE STRESS. The bioavailability of NO not only depends on its generation by eNOS but also is influenced by ROS-mediated breakdown. The low NO bioavailability is partly caused by the reaction of ROS with NO to form peroxynitrite. The application of laminar flow to intact vascular segments has been shown to increase ROS production for a short time period (117), with NAD(P)H being the major source (48). However, extended periods of CET result in a reduced expression of hypoxanthin (156) and NAD(P)H oxidase (2) and a stimulation of radical scavenging systems that include copper-zinc-containing superoxide dismutase (97), extracellular superoxide dismutase (69), and glutathione peroxidase (199). Another enzyme generating ROS in the vascular system is eNOS itself. Under a number of pathological conditions, the enzymatic reduction of molecular oxygen by eNOS is no longer coupled to L-arginine oxidation, resulting in ROS production (168, 213, 230). NO synthase uncoupling has been implicated in several pathologies including atherosclerosis (86), diabetes (114), and CHF (232). A critical factor for NO synthase uncoupling is the bioavailability of tetrahydrobiopterin (BH_4), a cofactor for the enzymatic reaction (9). Polymorphisms in the *GCH1* gene, which code for GTP-cyclohydrolase I, the rate-limiting enzyme in the biosynthesis of BH_4, result in substantial reductions in both plasma and vascular BH_4 levels. This was associated with increased levels of ROS and a reduced vasorelaxation to ACh (14). Finally, cell culture experiments using endothelial cells have provided some evidence that elevated blood flow increases BH_4 levels (112, 222).

ENDOTHELIUM AND miRNA. As described for the skeletal muscle, endothelium-specific miRNAs, called "angiomiRs," are also expressed in the vascular tissue. As summarized in a recent review, Marin et al. (136) propose a miR network regulating redox status and endothelium function. For example, atheroprotective flow, as achieved by CET, modulates the expression of miR-221, miR-221, miR-21, miR-17, and others. This finally leads to an antioxidative environment via the modulation of oxidative and antioxidative enzymes resulting in better endothelial function. Cell culture experiments have provided evidence for the involvement of miR-21 in shear stress-induced activation of eNOS (220). Overexpression of miR-21 activated eNOS, resulting in a 3.7-fold higher NO production, whereas inhibition of miR-21 abolished this shear stress-mediated eNOS activation. Large-scale miRNA profiling in human umbilical vein endothelial cells identified miR-92a as an atheromiR candidate whose expression is preferentially upregulated by the combination of low shear stress and atherogenic oxidized LDL (128). The expression of eNOS is also influenced by miR-92a. The flow-induced transcription factors Krüppel-like factor (KLF)2 (211) and KLF4 (64), which are inducers of eNOS expression, were shown to be targeted by miR-92a. Furthermore, overexpression of miR-92a directly suppressed eNOS expression (228).

One of the most highly expressed miRNAs in endothelial cells is miR-126. The results of the current literature investigating the impact of shear stress on miR-126 expression are conflicting: miR-126 was not identified by miR expression profiling in endothelial cells exposed to either oscillatory or laminar shear stress (153), but this contrasts with a recent report by Mondadori dos Santos et al. (147), where exposure of human umbilical vein endothelial cells to laminar shear stress for 24 h resulted in a significant ~2-fold increase in miR-126 expression.

Myocardium. Thirty to forty years ago, when exercise training was proposed to patients with CHF, the biggest fear of the clinicians was that the increased load would lead to acute injury and even sudden cardiac death. That this fear was unsubstantiated was first documented by Sullivan et al. (196). In their small study, they documented that 4–6 mo of CET did not deteriorate LVEF and even tended to improve cardiac output. Twelve years later, this result could be confirmed by the first large prospective study performed by Hambrecht et al. (79), which demonstrated an improvement of LVEF by 30–35% and a reduction in left ventricular end-diastolic diameter. A recently published meta-analysis incorporating 15 randomized controlled trials with 813 patients also concluded that CET reverses left ventricular remodeling and improves LVEF and end-diastolic and end-systolic volume (36). However, the mode of exercise as well as the duration have an impact on LVEF improvement. In the meta-analysis from Chen et al. (36), aerobic training demonstrated a significant improvement in LVEF, whereas patients performing RT or RT combined with CET did not exhibit such benefit. In addition, the benefit in LVEF was greater if CET was performed for >6 mo (36).

What are the mechanisms leading to these beneficial effects on the myocardium? Because of the absence of myocardial biopsies for molecular analysis before and after participating in an exercise training program, one can only speculate that the observed myocardial effects are mainly secondary to an afterload reduction and an improvement in resting blood pressure (71, 80) due to better endothelial function (76, 77). However, studies analyzing effects of exercise training in animal models of CHF and ischemia-reperfusion have revealed that molecular changes also occur in the myocardium finally leading to myocardial remodeling including physiological myocardial hypertrophy (61, 144, 235). Apart from the known effects of exercise training on the neurohormonal system, at least three further mechanisms have to be mentioned: *1*) the modulation of the catabolic/anabolic system, *2*) alterations in Ca^{2+} handling, and *3*) the modulation of stem cell proliferation.

MODULATION OF THE CATABOLIC SYSTEM. As already described for the skeletal muscle system, exercise training also influences the UPS in the myocardium. When CHF animals were analyzed, a significant upregulation and activation of the UPS and myostatin were detected. After these mice were exercised on a treadmill for 4 wk, these alterations were significantly reduced (1, 122). In addition, exercise training also induced the activation of IGF-I, leading to receptor-mediated activation of the PI3K/Akt signaling pathway resulting in myocardial hypertrophy [reviewed by Wei et al. (221)]. There seems also to be a cross talk between the ubiquitin E3 ligase MuRF-1 (component of the UPS) and IGF-I because overexpression of MuRF1 abolished IGF-I-induced cardiac hypertrophy (217).

MODULATION OF CA^{2+} HANDLING. Myocytes isolated from animals exhibiting heart failure are characterized by altered excitation-contraction coupling (ECC). One explanation for this observation is a diminished Ca^{2+} uptake rate into the sarcoplasmic reticulum due to reduced activity of sarco(endo) plasmic reticulum Ca^{2+}-ATPase (SERCA) (83, 129, 172). Additionally, it has been reported that in CHF, Ca^{2+} leak via ryanodine receptors (RyRs) increases, further limiting sarcoplasmic reticulum Ca^{2+} concentration (7, 138). In a rat model of CHF, RyR2 exhibited lower levels of *S*-nitrosylation, associated with impaired Ca^{2+} handling and contractility, the consequence of an imbalance between ROS and reactive nitrogen (201). Is exercise training able to modulate this altered ECC in CHF? When mouse and dog models of CHF were analyzed, an exercise-training intervention (CET for 8 wk) was sufficient to attenuate the alterations in Ca^{2+}-handling proteins (129, 172). In addition, HIIT activates Ca^{2+}/calmodulin-dependent protein kinase II, leading to a hyperphosphorylation of phospholamban (104), which in its phosphorylated form no longer inhibits SERCA2a. In conjunction with an increased expression of the Na^+/Ca^{2+} exchanger (225), this leads to improved Ca^{2+} cycling and finally to better cardiomyocyte contractility.

MODULATION OF STEM CELL PROLIFERATION. Over the last decades, the paradigm that the heart is a postmitotic organ without any capacity for self-renewal has changed. Small primitive cells [endogenous cardiac stem cells (eCSCs)], positive for the stem cell factor c-Kit and negative for hematopoietic lineage markers, have been detected using immunohistochemical methods or FACS in rodents, dogs, and human biopsies [reviewed by Hesse et al. (89)]. These stem cells are self-renewing, clonogenic, and multipotent. The presence of newly formed cardiomyocytes in normal and pathological hearts was further supported by bromodeoxyuridine (BrdU) staining (126) and carbon-14 dating (25). Using the carbon-14 method, a rate of cardiomyocyte renewal of 1 and 0.45%/yr was reported at an age of 25 and 75 yr, respectively (25). This would suggest that ~50% of the myocytes are replaced during a normal lifespan.

Does exercise training have any influence on cardiac stem cell activation and the renewal capacity of these cells? It is well documented that the adult heart responds to increased workload with cardiac stem cell activation (65, 219). In an elegant study, Waring et al. (219) exercised male Wistar rats on a treadmill using either CET (55–60% $\dot{V}O_{2max}$, 30 min/day, 4 days/wk for up to 4 wk) or a high-intensity protocol (85–90% $\dot{V}O_{2max}$, 30 min/day, 4 days/wk for up to 4 wk) and quantified eCSCs by immunohistochemistry and newly formed cardiomyocytes by BrdU staining. After 4 wk of exercise training, the number of newly formed cardiomyocytes (double positive for BrdU and α-sarcomeric actin) progressively increased with the highest number detected in high-intensity training compared with CET and sedentary rats. In addition, capillary density also increased in exercising animals, thereby balancing the generation of new myocytes and angiogenesis. This observation was confirmed in another animal study (231), where swim training of C57BL/6 mice also resulted in a significant increase in c-Kit-positive stem cells. This upregulation was time dependent and detectable in the left and right ventricles of the trained mice. At the molecular level, it has been speculated

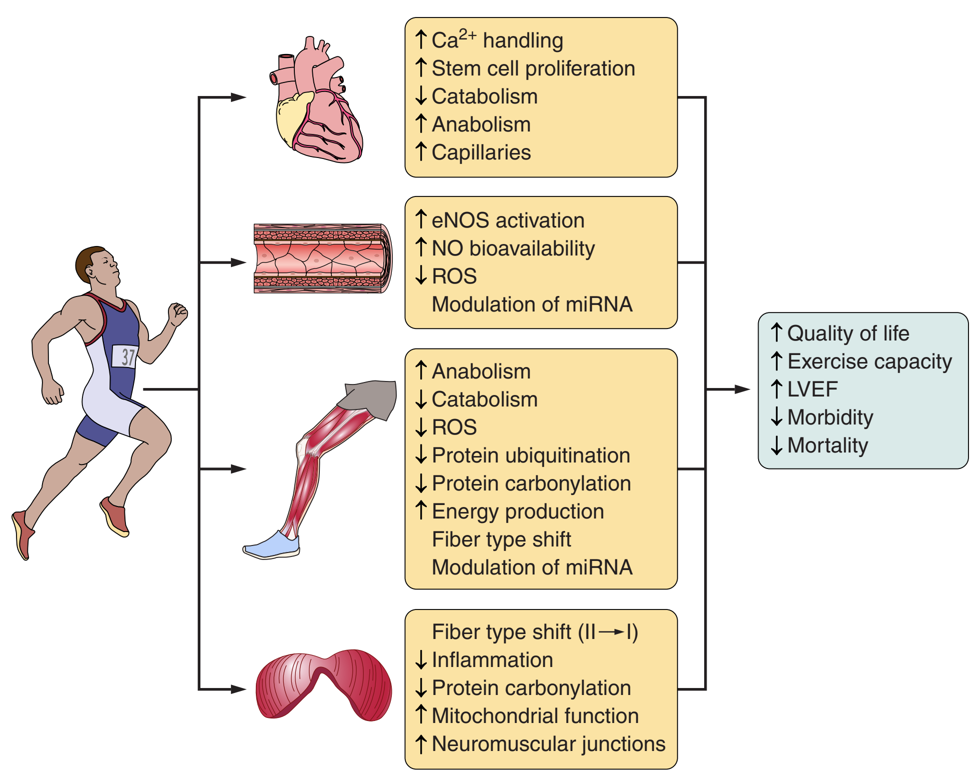

Fig. 5. Summary of molecular effects that exercise training elicits on the heart, endothelium, skeletal muscle, and diaphragm finally leading to improvement spanning from quality of life to exercise capacity and morbidity and mortality.

that the reduction of the transcription factor CCAAT/enhancer-binding protein-3β and increased expression of CBP/p300-interacting transactivator with ED-rich carboxy-terminal domain 4 (CITED4) are central signals for cellular renewal by eCSCs (28).

Conclusions

All of the above-described training methods have beneficial effects on both patients and healthy subjects. During the last years, we have also come to understand molecular changes in the skeletal muscle, endothelium, and myocardium responsible for the exercise intolerance often observed in HF patients and the impact of exercise training. By unraveling the molecular changes in different organ systems, it also became evident that exercise training evokes its beneficial effect in a multifactorial way (Fig. 5). As a result, exercise training leads to a panoply of positive effects spanning, among many others, from improved quality of life to reduced morbidity and mortality.

Realistically, however, the question is not so much which exercise protocol is superior or what molecular changes occur, but rather how we can design training programs that will be attractive to a large proportion of healthy subjects and patients alike. Offering not only attractive but also effective training programs to patients is the key to improved compliance and a lifelong increase in physical activity.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

V.A. and J.N. prepared figures; V.A., B.R., M.U., and J.N. drafted manuscript; V.A., B.R., M.U., and J.N. edited and revised manuscript; V.A., B.R., M.U., and J.N. approved final version of manuscript.

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